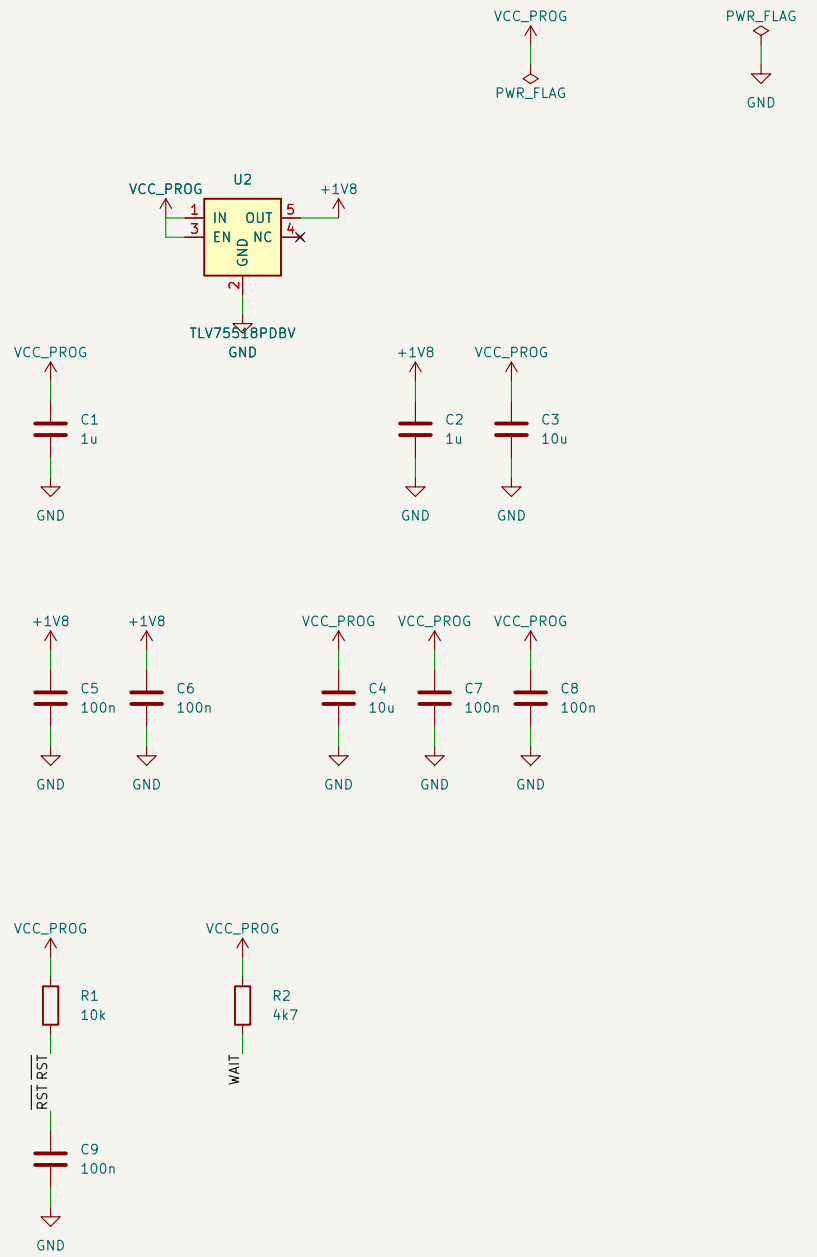
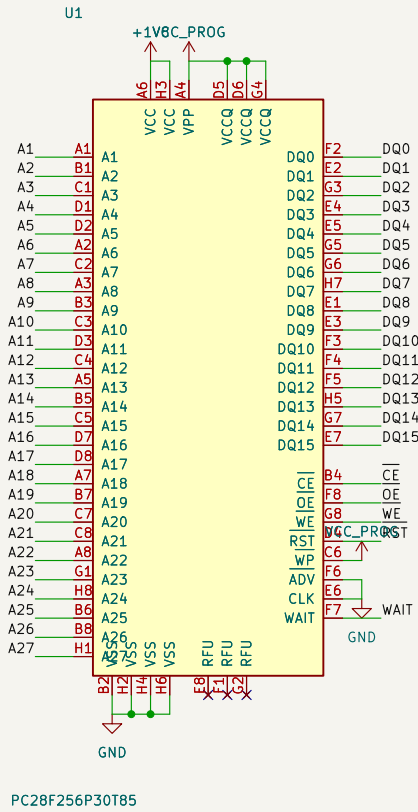
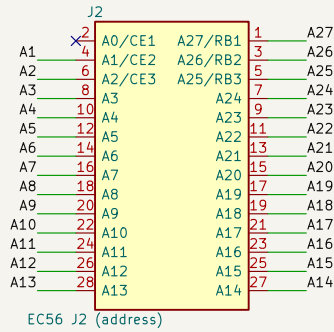
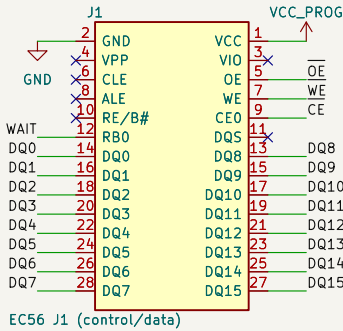


FlashCAT (FlashcatUSB Mach1 / XPORT) adapter for Intel/Micron P30 StrataFlash in 64–ball Easy BGA (e.g. PC28F256P30T85 / RC28F256P30T85), using an HMILU BGA64–1.0–TP21NS 10x13 mm BGA socket.



- NOTES
- Ball names follow the P30 datasheet (row letter A–H along the 13 mm side, column number 1–8). Pads of the socket footprint carry the same names; mapping remapped from the S29 footprint supplied by Embedded Computers.
 - x16 WORD addressing: FlashcatUSB drives header A1..A27 in x16 mode, so P30 A1 (LSB) → header A1. Header A0 is unused. A25 is used by 512 Mbit parts; B8 = A26 (1 Gbit) and H1 = A27 (2 Gbit) only exist on Micron P30–65nm (RFU on 130 nm parts), wired as EC’s TSOP–56 Type–D does.
 - P30 VCC (core) is 1.7–2.0 V, ABS MAX 2.5 V. U2 makes 1.8 V from the programmer VCC: programmer at 3.3 V → regulated 1.8 V; programmer at 1.8 V → U2 in dropout, ~1.78 V. VCCQ (I/O, 1.7–3.6 V) is fed from header VCC, header VIO is unused (same as the EC P30 TSOP–56 Type–D adapter). VPP (VPPL 0.9–3.6 V) is tied to header VCC; U2 feeds VCC only.
 - ADV# and CLK tied to GND (asynchronous mode). WAIT → header RB0 with 4.7k pull–up. RFU open.
 - RST#, WP# are not on the EC header. WP# is tied to header VCC (lock–down off, as the EC TSOP–56 Type–D adapter); RST# is pulled up to header VCC with a ~1 ms power–on delay. CE#, OE#, WE# are driven directly by the programmer (no pull–ups, as the EC adapters).
 - Unused EC header pins (VPP, VIO, CLE, ALE, RE/B#, DQS, A0) are left open.

Plugs into the Embedded Computers 56–pin dual header (FlashcatUSB Mach1 / XPORT)
Socket: HMILU BGA64–1.0–TP21NS. Target: Intel/Micron StrataFlash P30 (Easy BGA–64)
brainstorm@nopcode.org

Sheet: /
File: flashcat_p30.kicad_sch

Title: FlashCAT P30 Easy BGA–64 adapter

Size: A3 Date: 2026–10–02 Rev: C
KiCad E.D.A. 10.0.6 Id: 1/1